

Advertising, Social Media and Parasocial Recommendations

New Influencer-Centered Modeling Approach

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Model Setup: The Product Landscape

- Multiple product categories
 - ⇒ denote $k \in \{A, B, \dots\}$
 - ⇒ will allow for influencer “specialization”
 - ⇒ relevant only for informational purposes

Not yet used right now

- Multiple quality dimensions
 - ⇒ denote $j \in \{1, 2, \dots\}$
 - ⇒ relevant for realized utility

Model Setup: The Firm Side

The Monopolist

- Single firm with high (H) or low (L) quality in each dimension [generically q_j]
- Realization of all q_j are private information with $P[q_j = H] \equiv \pi$
- Produced at constant, homogeneous marginal cost c
- Competes (indirectly) against outside option
- Can advertise product on social media at exogenous cost κ

The Outside Option

- Available instead of purchasing the product
 - ⇒ Can be thought of as background Bertrand competition with cost $c_0 = v_0 + p_0$

Model Setup: The Consumer Side

- One (representative) consumer with unit demand
- Randomly drawn product dimension type $\theta \in J$
- Receives utility v_0 from outside option and v_q from the new product
 - ⇒ utility of product given by $u_c(\theta) = v_{q_\theta}$ [i.e. quality of dimension θ]
 - ⇒ normalize $v_L = 0$, simplify notation $v_H = v$
- Can obtain **one** recommendation r prior to purchase
 - ⇒ either from environment (non-strategic) or social media (potent. strategic)

Assumption 1

The outside option v_0 is sufficiently large to dominate random purchase. That is, $v_0 > \pi v - p$.

Model Setup: The Influencer Side

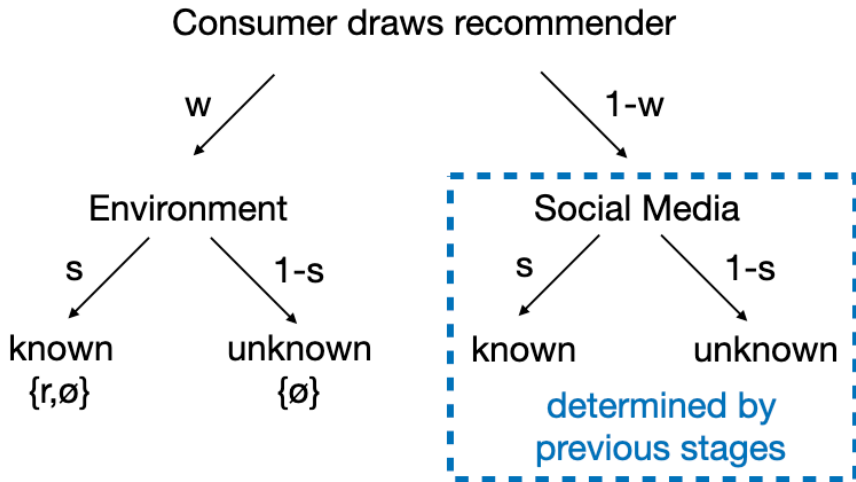
- One (representative) Influencer that can give recommendations
 - Product dimension type $\iota \in J$
 - direct utility from payment, indirect utility from value-added to consumer
 $\Rightarrow$ altruism parameter α captures trade-off, distr. following (cont.) F_α

$$u_{\mathcal{I}} = (1 - \alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - u_c(\emptyset)] = (1 - \alpha)\Pi_{\mathcal{I}} + \alpha[\mathbb{E}[v_\theta|r] - p - v_0]$$
 $\Rightarrow$ non-strategic environment encapsulated by $\alpha = 1$
- Type-match probability ("similarity") $\sigma \Rightarrow$ potentially σ_0, σ_I

Parasocial Relationships

The consumers parasocial relationship is captured by overestimating the influencer's similarity, i.e., $\sigma_\eta > \sigma$. [Specific form to be determined]

Model Setup: The Recommendation Procedure



Model Setup: Timing

1. Fundamentals are drawn ($p, \kappa, q, v_j, \dots$)
2. Firms learn q , knowing recommenders learn v_i
3. Firm decides on endorsement
4. Influencer decides on accepting endorsements
5. Consumer draws recommender
6. Consumer observes potential recommendation
 $\Rightarrow$ can distinguish genuine from paid recommendations
7. Consumer makes purchase decision
 $\Rightarrow$ utilities are realized

An Endorsement-Free Benchmark

- Outside option of $v_0 \Rightarrow$ by assumption, obtained without recommendation
- Consumer updates belief about v_θ after receiving a recommendation: $\pi \xrightarrow{r} \pi_r$

$$\pi_r \equiv \mathbb{P}[v_\theta = v | r] = \sigma \frac{1 \cdot \mathbb{P}[v]}{1 \cdot \mathbb{P}[v] + 0 \cdot \mathbb{P}[0]} + (1 - \sigma) \frac{\pi \cdot \mathbb{P}[v]}{\pi \cdot \mathbb{P}[v] + \pi \cdot \mathbb{P}[0]} = \sigma + (1 - \sigma)\pi$$

$\Rightarrow$ assume expected utility $\pi_r v - p = \pi v + (1 - \pi)\sigma v - p > v_0$

- without endorsements ($\Pi_{\mathcal{I}} = 0$), influencer's interests perfectly aligned
 $\Rightarrow$ recommend a (known) product iff $v_k = v$

Equilibrium of the Endorsement-Free Benchmark

Without endorsements, there is a unique equilibrium in which any high-value product gets recommended and all recommendations are followed.

The HML-Endorsement Equilibrium

HML-Endorsement Equilibrium

For sufficiently small κ there exists an equilibrium with endorsements in which

- (a) All firm types ($q = H, M, L$) decide to use endorsements at cost κ .
- (b) There exist thresholds $\bar{\alpha}_s$ and $\bar{\alpha}_{\bar{s}}$ such that an influencer
 - i. with $\alpha \leq \bar{\alpha}_s$ endorses the product unseen.
 - ii. with $\alpha \leq \bar{\alpha}_{\bar{s}}$ endorses any product and $\alpha > \bar{\alpha}_{\bar{s}}$ only $v_l = v$ products.
- (c) All consumers follow genuine recommendations.
- (d) Sufficiently parasocial consumers ($\sigma_\eta > \bar{\sigma}$) also follow endorsements.

- Equilibrium with full firm-type pooling

Firm Decision

- Firm endorses if costs are sufficiently small

$$\kappa \leq (D_0^q - D_E^q)(p - c)$$

⇒ effective demand with (D_E^q) and without (D_0^q) endorsements depends on v_l

	$v_l = v$	$v_l = 0$
D_0	s	0
D_E	$s + (1 - \omega)(1 - s)F_\alpha(\bar{\alpha}_s)$	$(1 - \omega)sF_\alpha(\bar{\alpha}_s) + (1 - \omega)(1 - s)F_\alpha(\bar{\alpha}_s)$

- Note that absolute gains $(D_0^q - D_E^q)$ decrease with q
 ⇒ all firms use endorsements as long as $\kappa \leq (1 - \omega)(1 - s)F_\alpha(\bar{\alpha}_s)(p - c)$

Consumer Decision

- Reminder: Consumers can differentiate paid from genuine recommendations
 $\Rightarrow$ paid ones can still be **aligned** (i.e., iff $v_\ell = v$) for high $\alpha \Rightarrow$ denote prob. ψ_α
- Genuine recommendations always aligned (Benchmark) $\Rightarrow$ always followed
- Follow paid recommendation iff v_0 or p sufficiently small and

$$\mathbb{E}[v_\theta | r^p] = \psi_\alpha(\pi + (1 - \pi)\sigma)v + (1 - \psi_\alpha)\pi v = [\pi + (1 - \pi)\sigma\psi_\alpha]v \geq v_0 + p$$

$$\Rightarrow \text{satisfied for sufficiently parasocial consumers} \Rightarrow \sigma \geq \frac{v_0 + p}{(1 - \pi)\psi_\alpha v} - \frac{\pi}{(1 - \pi)\psi_\alpha}$$

$\Rightarrow$ denote probability of following paid recommendations by ψ_η

Influencer Decision

- Recall influencer's utility: $u_{\mathcal{I}} = (1 - \alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - v_0]$
 - Three influencer types, accept endorsement iff
 - $\bar{s} : (1 - \alpha)\kappa + \alpha[\pi v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \pi v + p + v_0}$
 - $s|v : (1 - \alpha)\kappa \geq 0$
 $\Rightarrow$ always true (interests perfectly aligned)
 - $s|0 : (1 - \alpha)\kappa + \alpha[(1 - \sigma)\pi v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_s \equiv \frac{\kappa}{\kappa - (1 - \sigma)\pi v + p + v_0}$
- $\Rightarrow$ aligned paid recommendation probability $\psi_{\alpha} = \frac{1 - F_{\alpha}(\bar{\alpha}_s)}{1 + F_{\alpha}(\bar{\alpha}_{\bar{s}})}$

Non-Existence of L-Endorsement Equilibria

Non-Existence of L-Endorsement Equilibria

In the baseline, there cannot exist an equilibrium in which only the L -type uses endorsements while the other types rely solely on genuine recommendations.

- Equilibrium possible from firm side with sufficiently high κ
- Paid recommendations accepted by sufficiently selfish influencer (low $\bar{\alpha}$)
- paid recommendation credibly signals $q = L \Rightarrow v_\theta = 0$
 - $\Rightarrow$ no paid recommendation followed by consumers
 - $\Rightarrow$ no reason to endorse for L -type firm

What's ahead

- Influencer indirect utility conditional on following recommendation (status quo)?
- Continuous vs. binary parasocial state space?
- Characterize the prob. 0 mixed endorsement equilibria?
- ML-Endorsement Equilibrium
- Comparative Statics and Welfare/CS Analysis

- Introduce product categories and influencer “specialization” $\zeta \in K$
 $\Rightarrow$ affects σ and possibly ω_ζ $\rightarrow$ allows parasociality as spillover effects
- Endogenize ω (firm choice) $\rightarrow$ limited signalling
- Costly quality investigation for unknowing influencer

That's All Folks!

If you have questions, comments or feedback,
don't hesitate to contact me or stop by.

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Appendix: Influencer Decision with unconditional utility

- Recall influencer's utility: $u_{\mathcal{I}} = (1 - \alpha)\Pi_{\mathcal{I}} + \alpha[u_c(r) - v_0]$
 - Three influencer types, accept endorsement iff
 - $\bar{s} : (1 - \alpha)\kappa + \alpha\psi_{\eta}[\pi v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}} \equiv \frac{\kappa}{\kappa - \psi_{\eta}(\pi v - p - v_0)}$
 - $s|v : (1 - \alpha)\kappa + \alpha\psi_{\eta}[\sigma v + (1 - \sigma)\pi v - p - v_0] \geq \alpha[\sigma v + (1 - \sigma)\pi v - p - v_0]$
 $\Rightarrow \alpha \leq \bar{\alpha}_{\bar{s}}^v \equiv \frac{\kappa}{\kappa + (1 - \psi_{\eta})(\sigma v + (1 - \sigma)\pi v - p - v_0)}$
 - $s|0 : (1 - \alpha)\kappa + \alpha\psi_{\eta}[(1 - \sigma)\pi v - p - v_0] \geq 0$
 $\Rightarrow \alpha \leq \bar{\alpha}_s^0 \equiv \frac{\kappa}{\kappa - \psi_{\eta}((1 - \sigma)\pi v - p - v_0)}$
- $\Rightarrow$ one can show: $\bar{\alpha}_{\bar{s}}^v > \bar{\alpha}_{\bar{s}} > \bar{\alpha}_s^0$